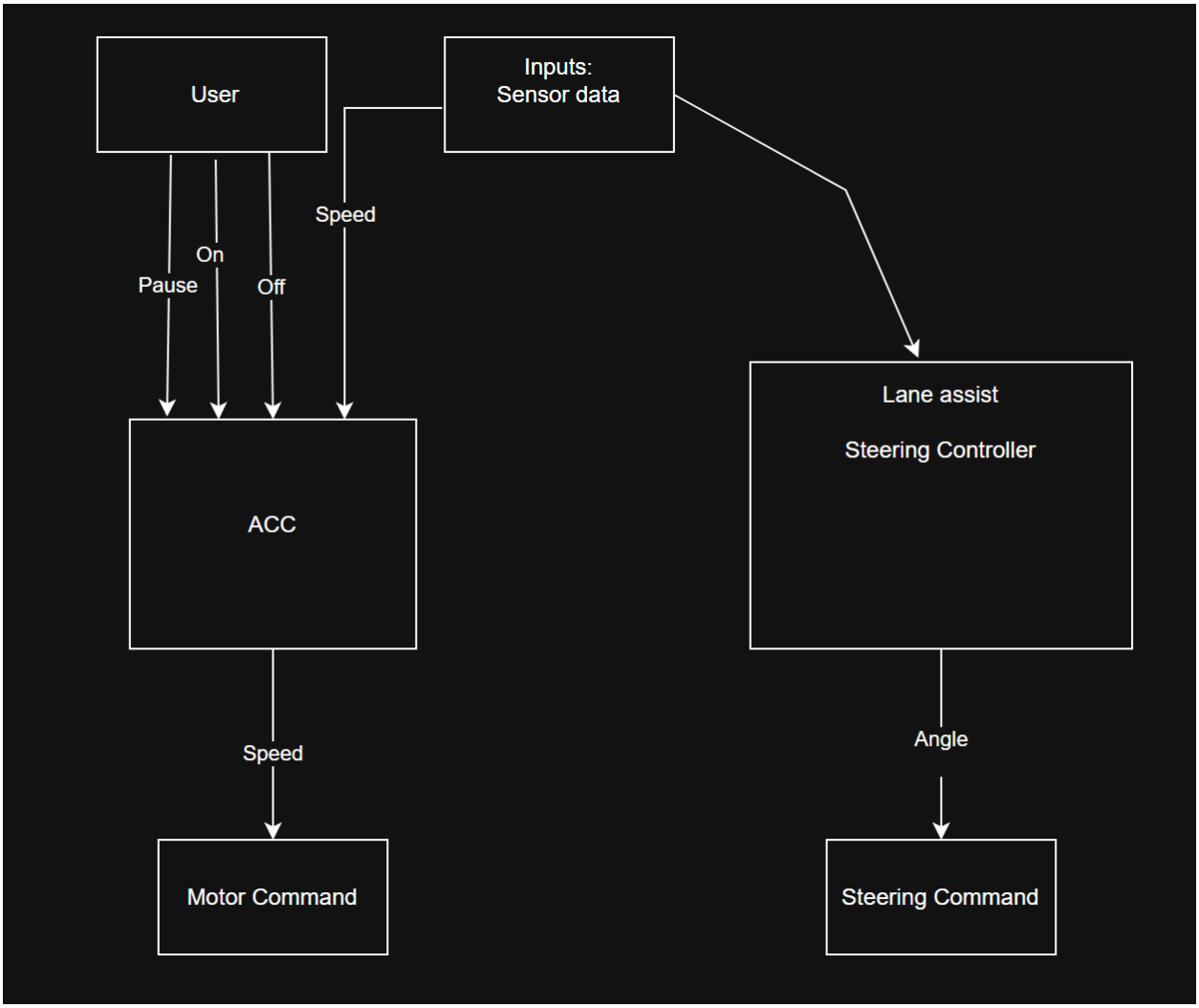
Autonomous vehicle control – Checkpoint 1

Names of team members and individual responsibilities:

- Kael Bjorndahl: Lane keeping assist (LKA)
- Mason Meinig: Lane keeping assist (LKA)
- Reagan Price: Adaptive Cruise Control (ACC)
- Holden Krueger: Adaptive Cruise Control (ACC)

Block Diagram:



Inputs req by subsystem

- Lane keeping assist:
    - Sensor measurements for lane distance from vehicle
    - Sensor measurements for current steer angle
    - Motion trajectory output
    - user input(on/off)

- ACC:
    - User inputs: (On, off, inc, dec, set speed)
    - Sensor measurements (current speed)

Outputs req by subsystem:

- Lane keeping assist:
    - Desired steer angle based motion trajectory output.

- ACC:
    - Actual speed

Simulink/Stateflow components that will be developed:

- One autonomous vehicle component consisting of communication between individual components:
    - ACC component
    - LKA component

Hardware required:

- No, hardware required for initial development, but will interface with hardware including motor, steering servos, and any other sensors handled by other teams.

Risks and Dependencies:

- Incorrect/Delayed sensors data from perception team (team1)
- Non-matching sensor outputs our controller doesn't expect
- Overshoot or delay in PI/PID controllers
- Any hardware limits (turning range, motor speed, actuator delay)
- Integration issues between hardware, Simulink, and Stateflow

Measurables:

1. The speed should stay within __ of the desired speed

2. Safety override controller responds within 0.1s of safety invariant being true

3. center of vehicle stays within ___ +/- inches of desired path or center lane

Communication:

- Team 1 - Communicate vehicle/sensor state data
- Team 3 - Communicate desired speed and trajectory
- Team 4 - Communicate actuator commands and any safety overrides